Harder Trigonometry Exam Questions 3

Total - 63 Marks (1 hour, 38 minutes)

1. (Penrith 2022 Q13c)

   The equation $I(t) = 4\sin 0.02t + 8\cos 0.02t$ measures the current, I amperes, in a particular circuit after t seconds.

   (i) Express the function in the form $I(t) = A\sin(at + b)$, for $0 \leq b \leq \frac{\pi}{2}$. **3**

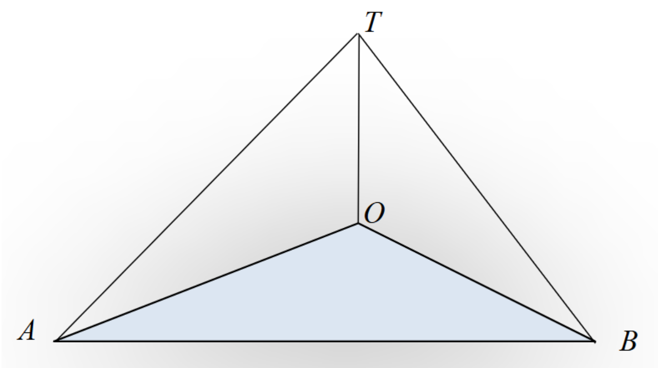
   (ii) When does the current first reach its maximum value? **3**

2. (Sydney Boys 2020 Q13b)

   Solve $\frac{\sin x + \cos x}{\sqrt{2}} = \frac{\sqrt{3}}{2}$ for $0° \leq x \leq 360°$. **3**

3. (Hornsby Girls 2023 Y11 Half-Yearly Q11b)

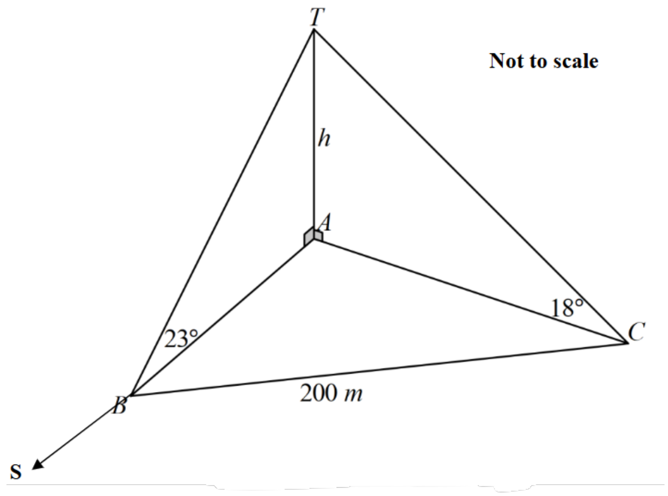
   Alex walks along a straight road on level ground. At one point, A, a tower is noticed on a bearing of 053°, with an angle of elevation of 21°. After Alex walks a further 230 m to a point B, the tower is on a bearing of 342° with an angle of elevation of 26°. Let the base of the tower be the point O and the top of the tower be the point T.



   (i) State the size of $\angle AOB$. **1**

   (ii) Find the height of the tower correct to the nearest metre. **3**

4. (Hornsby Girls 2024 Y11 Half-Yearly Q9d) TA is a vertical tower of height h metres. B and C are points at the same level as A, the foot of the tower, and $BC = 200$ m. The point B is due south of the tower and C is on a bearing of $120°T$ from the base of the tower. The angles of elevation to the top of the tower from B and C are $23°$ and $18°$, respectively. Find the height of the tower, correct to one decimal place. **4**

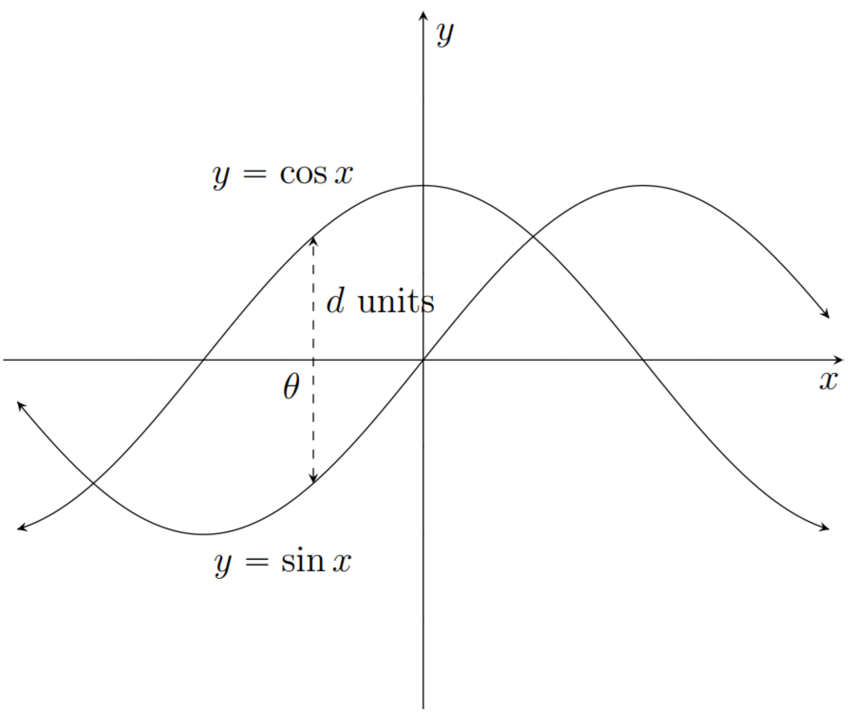


5. (Sydney Boys 2025 Q11c)

   Solve $\sin\theta + \cos\theta = -1$ for $0 \leq \theta \leq 2\pi$. **2**

6. (Sydney Girls 2025 Q13d)

   The graphs of $y = \cos x$ and $y = \sin x$ are sketched below.



   The dashed line indicates the greatest vertical distance, d units, between the two curves.

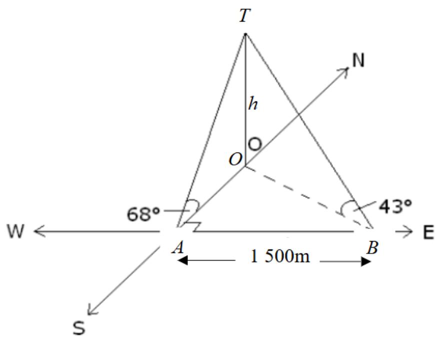
   The x-coordinate of the dashed line is θ.

   (i) Find d, giving reasons for your answer. **1**

(ii) Find θ, giving reasons for your answer. **1**

7. (Blacktown Boys 2022 Q24)
   Prove that $\tan\theta(1+\tan\theta)+(1+\cot\theta)=\frac{\tan\theta+1}{\sin\theta\cos\theta}$ **3**

8. (Hornsby Girls 2021 Y11 Half-Yearly Q11b) A lighthouse of height h metres, is observed from two points, A and B, in the same horizontal plane as the base of the lighthouse. From point A, due south of the lighthouse, the angle of elevation to the top of the lighthouse is $68°$. From point B, due east of point A, the angle of elevation to the top of the lighthouse is $43°$. Points A and B are 1 500 metres apart.



(i) Show that $OA = h\tan 22°$ **2**
(ii) Show that $h=\frac{1\ 500}{\sqrt{\tan^2 47°-\tan^2 22°}}$ **3**
(iii) Find the height of the tower correct to the nearest metre. **1**

9. (Sydney Girls 2022 Q2b)
   (i) Prove that $(\sin\alpha-\cos\alpha)^2=1-\sin 2\alpha$. **2**
   (ii) Hence, or otherwise, find the exact value of $\sin 75°-\cos 75°$. Leave your answer with a rational denominator. **2**

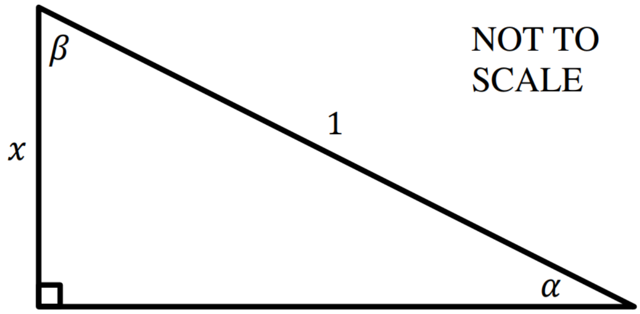
10. (Caringbah 2025 Q8b)
    (a) Find the exact value of $(\sin 22.5°+\cos 22.5°)^2$ **2**
    (b) If $\sin x=\frac{1}{3}$, $\cos y=-\frac{1}{\sqrt{3}}$ such that $\frac{\pi}{2}<x<\pi$ and $\pi<y<\frac{3\pi}{2}$. **4**
    Find the exact value of $\tan(x+y)$.
    (c) Prove that $\frac{\sin 3\theta}{\sin\theta}+\frac{\cos 3\theta}{\cos\theta}=4\cos 2\theta$ **3**

11. (Pymble 2019 Q30)
    Prove that $\frac{\cos x-\cos(x+2\theta)}{2\sin\theta}=\sin(x+\theta)$. **3**

12. (North Sydney Girls 2019 Q10c)
    The quadratic equation $ax^2+bx+c=0$ has roots $\tan\alpha$ and $\tan\beta$, where $\alpha<\beta$.

    Find a fully simplified expression involving a, b and c for $\tan(\alpha-\beta)$. **3**

13. (St George Girls 2020 Q12c)
    Consider the right-angled triangle below.
    i) Use the triangle to find an expression for α in terms of x. **1**



ii) By first finding a similar expression for β, solve $\sin^{-1} x - \cos^{-1} x = 0$. **3**
Give your answer in exact surd form.

14. (James Ruse 2024 Q9a)
(i) Using compound angle results, show that $\sin 3x = 3\sin x - 4\sin^3 x$. **3**

(ii) Consider the equation $\sin 2x = \sin 3x$. You are given that $x = \frac{\pi}{5}$ is a solution to this equation. Show that $x = \frac{3\pi}{5}$ is another solution to this equation. **1**

(iii) Hence, find the exact value of $\cos \frac{3\pi}{5}$. **3**

15. (James Ruse 2019 Q9a)
(i) If $t = \tan\theta$, show that $\tan 4\theta = \frac{4t(1-t^2)}{1-6t^2+t^4}$ **3**

(ii) Given the roots of $\tan 4\theta = \cot\theta$ are $\theta = \frac{\pi}{10}$ and $\theta = \frac{3\pi}{10}$. Find the exact value of $\tan \frac{\pi}{10}$. **3**